A Hierarchical Vision-Language Model-Guided Feature Fusion Framework for Referring Remote Sensing Image Segmentation

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***Abstract*—Referring remote sensing image segmentation (RSRIS) aims to achieve target-oriented, fine-grained understanding of geospatial information by leveraging both visual and linguistic modalities. Different from traditional remote sensing semantic segmentation, RSRIS needs to address more complex contextual relationships and pronounced scale variations inherent to remote sensing imagery, which pose significant challenges for precise alignment and fusion between textual semantics and visual features. To tackle these issues, we propose a novel framework, termed hierarchical vision-language model-guided feature fusion network (VSPNet), for vision-language guided remote sensing referring segmentation. Specifically, VSPNet adopts a hybrid backbone architecture based on convolutional neural network (CNN), Transformer, and Mamba models to extract rich, multidimensional visual representations, thereby enabling more effective cross-modal interaction. Furthermore, we design a hierarchical multimodal feature fusion strategy tailored for RSRIS: at the shallow feature level, a text-guided texture interaction module (TTIM) is introduced to enhance the integration of fine-grained texture details with textual cues; at the deep feature level, a text-guided semantic fusion module (TSFM) is developed to facilitate global contextual alignment between segmentation targets and semantic expressions in language. Extensive experiments on public benchmarks, RefSegRS and RSRIS dataset (RRSIS-D), demonstrate that VSPNet consistently outperforms state-of-the-art methods. Comprehensive ablation studies further verify the necessity and effectiveness of each constructed component. The code is available at https://github.com/NWPUFranklee/VSPNet.git**

***Index Terms*—Hierarchical feature extraction, referring image segmentation (RIS), remote sensing, vision-language fusion.**

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I. Introduction

REMOTE sensing semantic segmentation has become a cornerstone methodology for precise land cover analysis and comprehensive scene understanding [1], driven by advances in artificial intelligence and deep learning that have accelerated multimodal applications across diverse earth observation tasks [2], [3], [4], [5], [6]. The integration of heterogeneous modalities, such as optical imagery, thermal infrared, light detection and ranging (LiDAR), and synthetic aperture radar (SAR), has notably enhanced the robustness and richness of geospatial information [7], [8], [9], [10], [11], [12]. At the same time, the emergence of text-guided remote sensing tasks, including open-vocabulary semantic segmentation, has further broadened the scope of multimodal analysis [13], [14], [15], [16], [17]. Despite these developments, referring image segmentation in remote sensing, which leverages free-form natural language to guide pixel-level delineation, remains in its infancy. Different from traditional semantic or instance segmentation that maps each pixel to a predefined category, referring remote sensing image segmentation (RSRIS) identifies and segments objects corresponding to arbitrary textual descriptions, enabling more flexible and interactive geospatial analysis. This paradigm is particularly promising for applications such as environmental monitoring, disaster response, and military intelligence [18], [19], [20].

The origins of RSRIS can be traced to referring segmentation methods in natural scene images [21], [22], [23], where the fusion of computer vision and language introduces unique challenges in cross-modal semantic alignment and fine-grained localization. For instance, the locate-then-segment (LTS) framework [24] addresses limitations in reference object localization within natural scenes through a location-prioritization mechanism and lightweight segmentation module. Referring segmentation has also been explored in medical imaging, where the text-promptable propagation (TPP) strategy [25] utilizes cross-modal interactions and Transformer-based propagation for interactive text-guided segmentation in both 2-D and 3-D images. While medical images offer relatively fixed contextual structures and natural scenes emphasize diverse textures, both domains benefit from the alignment of textual semantics with visual content. However, remote sensing imagery exhibits fundamentally distinct characteristics, such as intricate spatial contexts, substantial scale variability, and modality-specific spectral properties, which are absent in other

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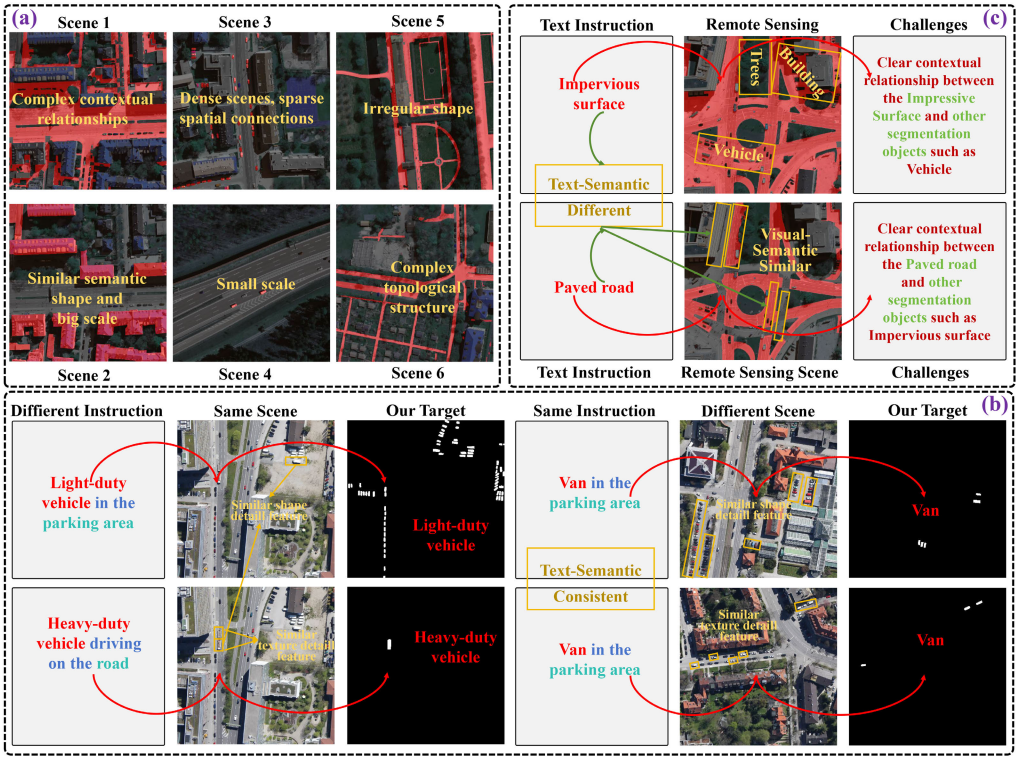


Fig. 1. Illustration of the three main challenges in remote sensing referring segmentation: (a) complex object distributions and diverse spatial contexts, (b) ambiguity in texture and semantics under different text instructions, and (c) difficulty in aligning global context and textual information.

domains. These aspects necessitate referential segmentation models explicitly designed for the complex demands of remote sensing. Recently, the construction of dedicated referring segmentation datasets [26], [27] has laid a solid foundation for progress in this field. Building on these resources, pioneering approaches have emerged. For example, BTDNet [28] addresses the visual-language gap and the challenge of distinguishing densely clustered objects with ambiguous boundaries via a bidirectional spatial correlation module and a dual-modal object learning strategy, achieving notable advances in segmentation precision. Nevertheless, as illustrated in Fig. 1, substantial challenges remain in remote sensing referring image segmentation (RIS), including robust cross-modal alignment and fine-grained object delineation in complex scenes.

1) *Limited Feature Representation:* Remote sensing imagery contains densely distributed and heterogeneous objects, often exhibiting wide spatial variance and weak interobject correlations [Fig. 1(a)]. Single-stream feature representation struggle to capture such fine-grained diversity in scale, shape, and spatial distribution.
2) *Insufficient Multilevel Interaction:* As shown in Fig. 1(b), textual instructions may refer to different objects with highly similar texture features within the same scene, or the same object may display differing visual details under varying textual queries. Robust alignment and fusion of texture-level features with semantic cues remains a critical unresolved issue.
3) *Inadequate Contextual and Semantic Fusion:* Fig. 1(c) highlights that remote sensing scenes often exhibit complex contextual relationships among specify objects and their environments. This discrepancy hinders the effective integration of global contextual information and textual semantics, challenging current frameworks that rely on text–image feature alignment.

These domain-specific challenges underscore the necessity for novel architectures capable of capturing fine-grained visual diversity, enhancing semantic–texture fusion, and leveraging global contextual information in a unified manner for remote sensing RIS.

To tackle the aforementioned challenges, we propose a novel remote sensing referring segmentation framework, termed hierarchical vision-language model-guided feature fusion network (VSPNet), built upon the MambaVision-CLIP architecture. Inspired by the MambaVision [29], VSPNet adopts a convolutional neural network (CNN)-mamba-transformer (CMTFE) hybrid paradigm for feature extraction, enabling hierarchical and multidimensional representations of remote sensing imagery. Specifically, our approach utilizes a shallow-to-deep feature extraction strategy to capture pyramid-shaped remote sensing features at multiple semantic levels. In VSPNet, features extracted by the CNN [30], Mamba [31], and Transformer [32] submodules each encode distinctive properties of the segmentation targets across different scales. For shallow CNN features, which emphasize visual texture details, we introduce a text-guided texture interaction module (TTIM) that leverages cross-attention to distinguish between similar and distinct texture patterns under textual guidance. For deeper semantic features, we propose a text-guided semantic fusion module (TSFM), which incorporates object-specific, prepositional, and nontarget information embedded in the textual descriptions to facilitate effective semantic fusion between visual and textual modalities. The main contributions of this work are summarized as follows.

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1) We propose VSPNet, a novel framework for remote sensing referring segmentation, which achieves state-of-the-art performance on public benchmarks, RefSegRS and RSRIS dataset (RRSIS-D), thus demonstrating its effectiveness and generalizability for referring segmentation task.
2) We design a hierarchical triple-branch feature extraction paradigm that integrates CNN, Transformer, and Mamba models, enabling rich multilevel feature representation and providing a robust foundation for subsequent text-image feature alignment and fusion.
3) We devise a TTIM based on cross-attention to enhance the alignment of shallow texture features with textual cues, thereby further improving the discriminative and robustness for fine-grained feature and enhancing the accurate positioning of objects with different scales.
4) We introduce a TSFM based on Mamba and Adapter, which deeply integrates textual and visual semantics by separately modeling target, positional, and nontarget information in text, significantly improving the modeling of contextual relationships in remote sensing scenes.

The article is structured as follows. Section II provides an overview of the related works. Section III introduces the proposed VSPNet framework and its associated components. The experimental results and discussion analysis are presented in Section IV. Lastly, the conclusion is drawn in Section V.

II. Related Works

A. Natural Scene RIS

Recent years have witnessed significant advances in RIS within natural scenes. Huang et al. [33] proposed DETRIS, which leverages densely interconnected low-rank visual feature propagation and a text adapter to enable parameter-efficient multimodal feature alignment with unaligned encoders. Ding et al. [34] introduced the Vision-Language Transformer, incorporating a query generation module for diverse language interpretations and a query balance module for adaptive query selection, thus addressing complex object relationships and the inherent variability of language expressions. To extend RIS beyond single-object scenarios, Liu et al. [35] proposed the generalized referring expression segmentation (GRES), which supports multiobject and no-object expressions through the RefCOCO dataset and the ReLA strategy. Shang et al. [36] presented Prompt-RIS, which employs cross-modal prompting and instance contrastive learning to bridge the cross-task gap between CLIP and SAM, and enhance cross-modal interaction. Nguyen-Truong et al. [37] proposed VATEX, introducing a CLIP Prior for query initialization and a Meaning Consistency Constraint to improve contextual coherence, thereby addressing the limited text understanding in prior RIS models. Yang et al. [38] addressed the underutilization of encoder-stage multimodal context with the language-aware visual Transformer, which incorporates a pixel-word attention module for enhanced cross-modal fusion. Wang et al. [39] developed an end-to-end RIS framework featuring a vision-language decoder for semantic transfer and a text-pixel contrastive learning strategy for robust alignment. Yang et al. [40] further introduced ReMamber, utilizing the Mamba Twister block to efficiently capture long-range dependencies with reduced computational overhead compared to Transformers. Extending RIS to the video domain, Liang and Zhang [41] proposed video referring segmentation framework, featuring the hierarchical temporal multimodal alignment module to exploit latent temporal information and interframe aggregation in referring video object segmentation. While Mamba-based architectures have shown promise in natural scene RIS, their application to remote sensing referring segmentation remains largely unexplored.

B. Referencing Remote Sensing Image Segmentation

Recent progress in referring image segmentation for remote sensing has focused on bridging the gap between fine-grained semantic alignment and the unique spatial complexities of remote sensing imagery. Chen et al. [42] introduced RSRefSeg, an early foundational model that leverages CLIP [43] to generate coarse text-visual alignment features, which are then converted into prompts for SAM [44], thus facilitating fine-grained semantic alignment. To further improve precision, Lei et al. [45] proposed FIANet, which decouples textual descriptions into object and spatial location cues and designs specialized modules for their fine-grained alignment and fusion with visual features. To address the challenges of varying spatial scales and rotation in remote sensing images, Liu et al. [46] developed the rotated multiscale interaction network (RMSIN), introducing the intrascale interaction module and cross-scale interaction module to enhance scale- and orientation-aware feature interactions. Ma et al. [47] tackled the issues of information loss and semantic forgetting in the decoding stage by proposing LSCF, which incorporates a sampling attention and a long-term semantic guidance decoder for improved feature fusion and long-term semantic retention. Further advancements have been made by Liu et al. [48] with CADFormer, which introduces the semantic mutual guidance alignment module (SMGAM) and text-enhanced cross-modal decoder (TCMD) to address coarse feature alignment and insufficient language feature utilization during decoding. Dong et al. [49] proposed CroBIM, featuring the context-aware prompt modulation and language-guided feature aggregation, specifically designed to handle complex geospatial relationships and diverse object scales in RSRIS. Beyond these task-specific approaches, recent advances in vision-language models (VLMs) tailored for remote sensing, such as GeoChat [50], RemoteCLIP [51], and EarthGPT [52], have demonstrated the potential of unified cross-modal representation learning in this domain. Despite these advances, challenges remain in achieving robust multilevel feature alignment and effective semantic fusion in remote sensing referring segmentation, motivating further research in this domain.

III. Methodology

In this section, we present the details of our proposed VSPNet framework for remote sensing referring segmentation, as depicted in Fig. 2. For visual feature representation, VSPNet adopts a novel asymptotic hierarchical feature extraction architecture, designed to capture fine-grained and domain-specific characteristics inherent in complex remote sensing

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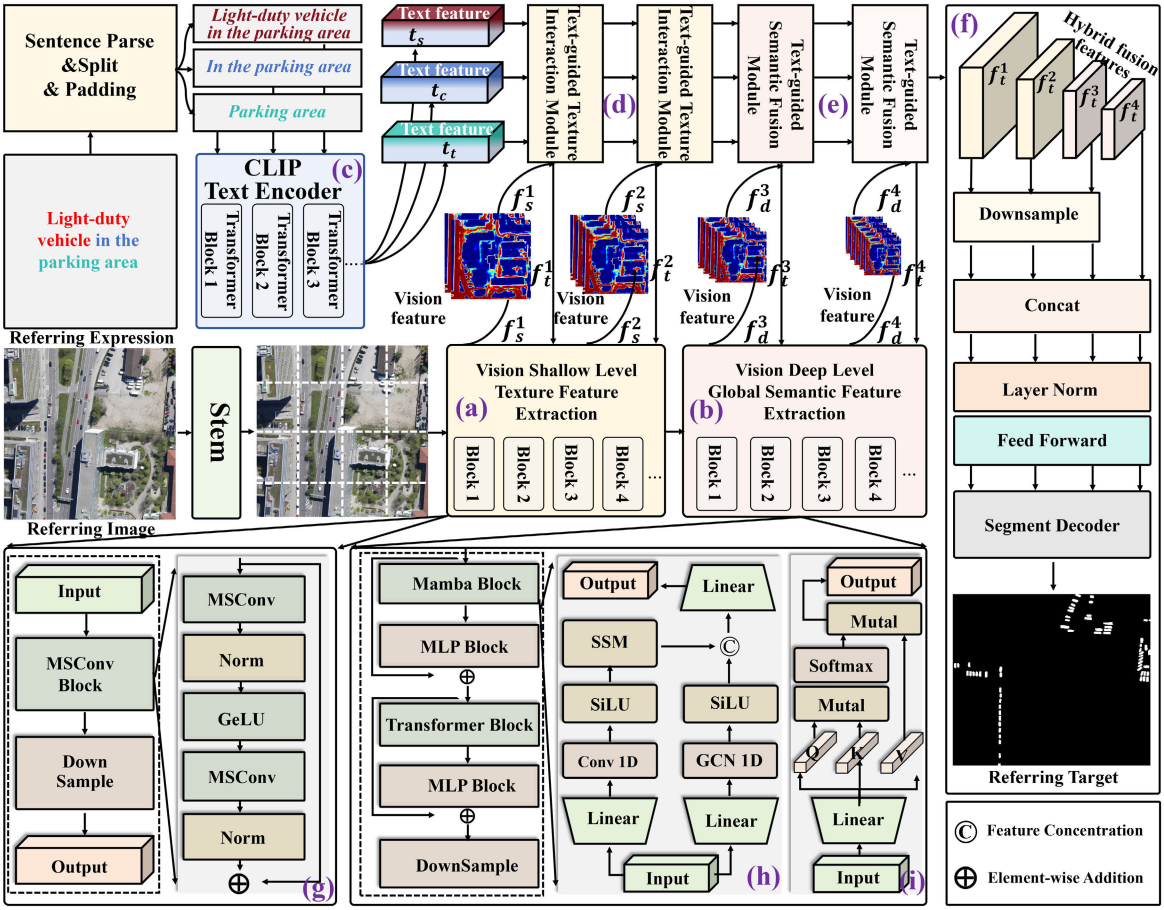


Fig. 2. Overview of the VSPNet architecture for remote sensing referring segmentation. The framework consists of four main components: (a) multiscale CMTFE architecture for capturing shallow texture and deep semantic visual features; (b) hierarchical visual feature extraction; (c) multibranch CLIP-based text encoding for original, spatial, and contextual instructions; (d) and (e) progressive text-guided texture interaction and semantic fusion modules for cross-modal alignment; and (f) multiscale hybrid decoder for generating the final segmentation mask.

imagery. This enables the model to learn rich, multilevel visual features that are crucial for accurate segmentation in challenging scenarios. For text feature representation, we utilize a pretrained CLIP text encoder to extract precise semantic embeddings from the referring expressions, ensuring robust and discriminative textual representation. To effectively bridge the semantic gap between the visual and textual modalities, VSPNet incorporates two key modules: a TTIM, which facilitates fine-grained fusion between textual cues and shallow visual features, and a TSFM, which progressively aligns and integrates semantic information across both modalities at deeper feature levels. These components jointly enhance the model's ability to perform accurate and context-aware referring segmentation in remote sensing scenes.

A. Overall Structure

VSPNet abandons the conventional feature extraction paradigm. Instead, drawing on the concept of VMamba [29], we propose the CMTFE feature extraction architecture, specifically tailored for remote sensing referring segmentation. As shown in Fig. 2(a), the CMTFE first employs a shallow level texture feature extraction module (SLTFEM), which is built upon a multiscale convolutional framework, to extract low-level texture features $\boldsymbol{f}_s^i \in \mathbb{R}^{C_i \times (H \times W/2^{i+1})}$ from the input image $\boldsymbol{I} \in \mathbb{R}^{3 \times H \times W}$, where $i \in \{1,2,3,4\}$ denotes the layer index and C_i the corresponding number of channels. These features capture shape, color, and fine-grained texture information. Subsequently, as shown in Fig. 2(b), the feature $\boldsymbol{f}_s^i$ is fed into the deep level semantic feature extraction module (DLSFEM) to obtain deep semantic features $\boldsymbol{f}_d^i$, which encode contextual relationships and higher-level semantics

$$\boldsymbol{f}_s^i = \varphi_{\text{SLTFEM}}(\boldsymbol{I}), \quad i \in \{1,2\} \tag{1}$$

$$\boldsymbol{f}_d^i = \varphi_{\text{DLSFEM}}\left(\boldsymbol{f}_s^2\right), \quad i \in \{3,4\}. \tag{2}$$

For the text modality, as depicted in Fig. 2(c), we parse the referring expression $\boldsymbol{T}_t$ using NLTK [53] to obtain a triplet of instructions: the original text ($\boldsymbol{T}_t$), the spatial relationship text ($\boldsymbol{T}_s$), and the contextual object text ($\boldsymbol{T}_c$). Each is encoded by a pretrained CLIP text encoder, yielding the features $\boldsymbol{t}_t, \boldsymbol{t}_s, \boldsymbol{t}_c$

$$\boldsymbol{T}_t, \boldsymbol{T}_s, \boldsymbol{T}_c = \varphi_{\text{NLTK}}(\boldsymbol{T}_t) \tag{3}$$

$$\boldsymbol{t}_k = \varphi_{\text{CLIP}}(\boldsymbol{T}_k), \quad k \in \{t, s, c\}. \tag{4}$$

After obtaining multilevel visual and textual features, we adopt a progressive fusion paradigm inspired by human cognition: first filtering candidate regions based on appearance, then refining localization by leveraging context and spatial cues.

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Algorithm 1 Pseudocode of VSPNet in a PyTorch-Like Style

```
# I: input remote sensing image; T_t: referring expression
# SLTFEM: Shallow Level Texture Feature Extraction Module
# DLSFEM: Deep Level Semantic Feature Extraction Module
# TTIM: Text-guided Texture Interaction Module
# TSFM: Text-guided Semantic Fusion Module
# Out: referring segmentation result # —Parse and encode the text—
T_t, T_s, T_c = NLTK_Parse(T_t)
t_t = CLIP_Text_Encoder(T_t)
t_s = CLIP_Text_Encoder(T_s)
t_c = CLIP_Text_Encoder(T_c) # —Extract multilevel visual features—
f_s_1 = SLTFEM_Block1(I)
f_s_2 = SLTFEM_Block2(f_s_1)
f_d_3 = DLSFEM_Block1(f_s_2)
f_d_4 = DLSFEM_Block2(f_d_3) # — Progressive cross-modal fusion —
f_t_1 = TTIM(t_t, t_s, t_c, f_s_1)
f_t_2 = TTIM(t_t, t_s, t_c, f_s_2)
f_t_3 = TSFM(t_t, t_s, t_c, f_d_3)
f_t_4 = TSFM(t_t, t_s, t_c, f_d_4) # — Multiscale fusion and decoding —
F_cat = Concat(f_t_1, f_t_2, f_t_3, f_t_4)
Out = Segment_Decoder(F_cat)
```

Specifically, we introduce two dedicated fusion modules: the TTIM as $\varphi_{\mathrm{TIIM}}(\cdot)$ and the TSFM as $\varphi_{\mathrm{TSFM}}(\cdot)$, as shown in Fig. 2(d) and (e). These modules sequentially integrate textual and visual features to yield fused representations $\boldsymbol{f}_t^i$ as follows:

$$\boldsymbol{f}_t^i = \varphi_{\mathrm{TIIM}}\left(\boldsymbol{t}_t, \boldsymbol{t}_s, \boldsymbol{t}_c, \boldsymbol{f}_s^i\right), \quad i \in \{1,2\} \tag{5}$$

$$\boldsymbol{f}_t^i = \varphi_{\mathrm{TSFM}}\left(\boldsymbol{t}_t, \boldsymbol{t}_s, \boldsymbol{t}_c, \boldsymbol{f}_d^i\right), \quad i \in \{3,4\}. \tag{6}$$

Finally, as depicted in Fig. 2(f), the decoder Φ_{Decoder} preserves multiscale adaptability between textual and visual features while maintaining a lightweight architecture suitable for efficient inference, consistent with the design principles of existing decoder [45]. The decoding process is as follows:

$$R = \Phi_{\mathrm{Decoder}}\left(f_t^i\right), \quad i \in \{1,2,3,4\} \tag{7}$$

where R denotes the final segmentation result. Algorithm 1 provides the pseudocode of VSPNet in a PyTorch-like style.

B. CMTFE Feature Extraction

To comprehensively capture both local texture and global semantic information from remote sensing images, we design a feature extraction architecture that integrates local texture encoding and global context modeling. This hybrid design enables the model to fully represent the multiscale characteristics of remote sensing data. Initially, as shown in Fig. 2(g), the input feature $\boldsymbol{X}_{in}$ is processed by a multiscale convolutional module $\varphi_{\mathrm{MSC}}(\cdot)$, which employs convolutional kernels of varying sizes to extract rich texture details at different spatial resolutions. This is followed by LayerNorm normalization $\varphi_{\mathrm{LN}}(\cdot)$, a GeLU activation $\varphi_{\mathrm{GeLU}}(\cdot)$, and a residual connection. The specific calculation is as follows:

$$\boldsymbol{X}_{out} = \varphi_{\mathrm{MSC}}\left(\varphi_{\mathrm{GeLU}}\left(\varphi_{\mathrm{LN}}\left(\varphi_{\mathrm{MSC}}\left(\boldsymbol{X}_{in}\right)\right)\right)\right) + \boldsymbol{X}_{in}. \tag{8}$$

This multiscale convolutional design enables the model to efficiently encode shallow texture features at multiple scales. To further enhance representation, the architecture adopts a global modeling stage that alternates between Mamba and Transformer (Self-Attention) substructures, as illustrated in Fig. 2(h) and (i). In each encoder stage, a Mamba block is followed by a Transformer block, and this sequence is repeated twice per stage, resulting in two hybrid units. The channel dimensions of both substructures are aligned, with 1×1 convolutions applied when necessary to ensure efficient feature fusion. Within each substructure, the input is first normalized using LayerNorm, followed by context modeling as $\varphi_{\mathrm{CT}}(\cdot)$, yielding an intermediate feature $\boldsymbol{Z}$

$$\boldsymbol{Z} = \varphi_{\mathrm{CT}}\left(\varphi_{\mathrm{LN}}\left(\boldsymbol{X}_{in}\right)\right) + \boldsymbol{X}_{in}. \tag{9}$$

An MLP layer is then applied to the normalized intermediate feature to produce the final output

$$\boldsymbol{X}_{out} = \varphi_{\mathrm{MLP}}\left(\varphi_{\mathrm{LN}}\left(\boldsymbol{Z}\right)\right) + \boldsymbol{Z}. \tag{10}$$

For spatial modeling, the causal convolution in the Mamba branch is replaced with a 1-D convolution $\varphi_{\mathrm{Conv}}(\cdot)$ to better capture local spatial relationships. To address the sequence bias and limited global context in SSM [54], a parallel branch is introduced using graph convolution $\varphi_{\mathrm{GCN}}(\cdot)$ and the SiLU activation $\varphi_{\mathrm{SiLU}}(\cdot)$ to capture complex topological dependencies. The outputs of both branches are concatenated and projected as follows:

$$\boldsymbol{X}_1 = \varphi_{\mathrm{SSM}}\left(\varphi_{\mathrm{SiLU}}\left(\varphi_{\mathrm{Conv}}\left(\boldsymbol{W}_{in}^1 \boldsymbol{X}_{in}\right)\right)\right) + \boldsymbol{X}_{in} \tag{11}$$

$$\boldsymbol{X}_2 = \varphi_{\mathrm{SiLU}}\left(\varphi_{\mathrm{GCN}}\left(\boldsymbol{W}_{in}^2 \boldsymbol{X}_{in}\right)\right) \tag{12}$$

$$\boldsymbol{X}_{out} = \boldsymbol{W}_{in}^3 \mathrm{Cat}\left[\boldsymbol{X}_1, \boldsymbol{X}_2\right] \tag{13}$$

where $\boldsymbol{W}_{in}^1$, $\boldsymbol{W}_{in}^2$, and $\boldsymbol{W}_{in}^3$ are learnable parameters, and $\boldsymbol{X}_1$, $\boldsymbol{X}_2$ denote the outputs of the SSM and GCN branches, respectively. The SSM-GCN concatenation integrates sequential and global context, alleviating sequence bias and enriching features, while maintaining the linear complexity and efficiency of the original Mamba design. The Transformer substructure follows the standard self-attention formulation:

$$\mathbf{Att}\left(\boldsymbol{X}\right) = \mathbf{S}\left(\frac{\boldsymbol{W}_Q \boldsymbol{X}_Q \cdot \left(\boldsymbol{W}_K \boldsymbol{X}_K\right)^{\top}}{\sqrt{d}}\right) \boldsymbol{W}_V \boldsymbol{X}_V \tag{14}$$

where $\mathbf{S}(\cdot)$ is the Softmax function and $\boldsymbol{W}_Q$, $\boldsymbol{W}_K$, $\boldsymbol{W}_V$ are learnable projection matrices.

C. Text-Guided Texture Interaction Module

To bridge the semantic gap between language and vision at the early feature level, the TTIM, as shown in Fig. 3(a), is designed to facilitate interactive fusion between language semantics and shallow visual features $\boldsymbol{X}_{in}$. First, we extract channel-wise and spatial saliency representations, $\boldsymbol{X}_{in}^m$ and $\boldsymbol{X}_{in}^a$, via Max and Maxpool operations, respectively. These are concatenated and projected with learnable weights $\boldsymbol{W}_{in}^4$ to obtain a

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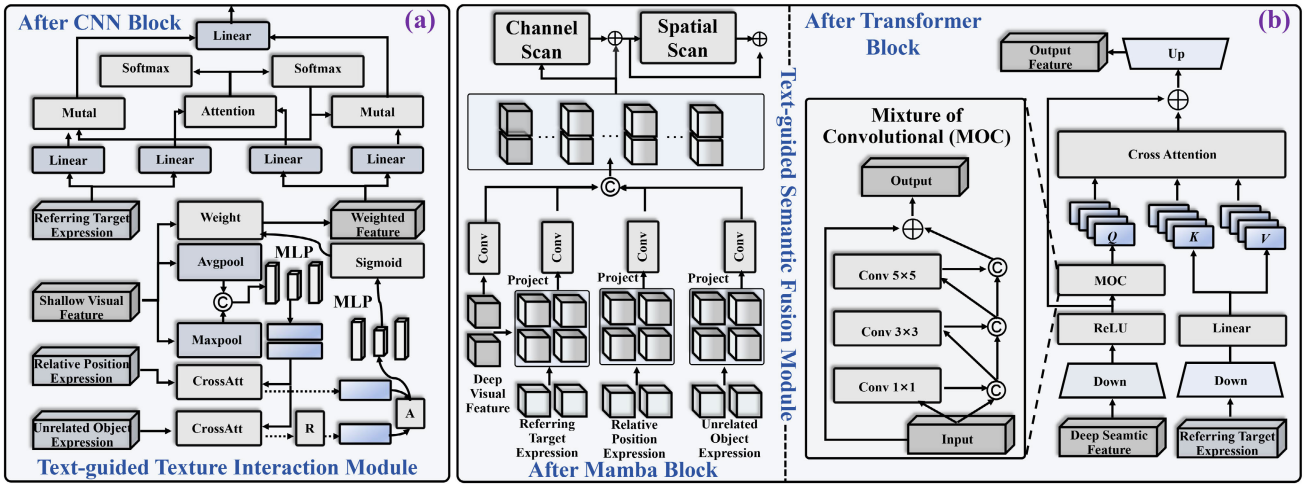


Fig. 3. Architecture of the proposed text-guided semantic interaction and fusion modules. (a) TTIM aligns shallow visual features with multiple types of textual expressions using mutual and cross-attention mechanisms after the CNN block. (b) TSFM integrates deep visual and textual semantics via multidimensional scanning, a MOC, and cross-attention operations after the Mamba and Transformer blocks, enabling comprehensive and adaptive multimodal feature fusion for remote sensing tasks.

compact hybrid representation $\boldsymbol{X}_{in}^{ma}$, which effectively reduces feature dimensionality of subsequent attention mechanisms

$$\boldsymbol{X}_{in}^{m}, \boldsymbol{X}_{in}^{a} = \mathrm{Max}\left(\boldsymbol{X}_{in}\right), \mathrm{Maxpool}\left(\boldsymbol{X}_{in}\right) \tag{15}$$

$$\boldsymbol{X}_{in}^{ma} = \boldsymbol{W}_{in}^{4}\mathrm{Cat}\left[\boldsymbol{X}_{in}^{m}, \boldsymbol{X}_{in}^{a}\right]. \tag{16}$$

The vision-language interaction in TTIM comprises two stages. In the first stage, we compute compensatory weights for visual feature refinement by modeling their relationship with two types of textual features. Specifically, we employ cross-attention to capture the interaction between $\boldsymbol{X}_{in}^{ma}$ and both the spatial position text feature $\boldsymbol{t}_s$ and the irrelevant object text feature $\boldsymbol{t}_c$, producing $\boldsymbol{XT}_{s}^{ma}$ and $\boldsymbol{XT}_{c}^{ma}$, respectively,

$$\boldsymbol{XT}_{k}^{ma} = \mathrm{Att}\left(\boldsymbol{X}_{in}^{ma}, \boldsymbol{t}_k, \boldsymbol{t}_k\right), \quad k \in \{s, c\}. \tag{17}$$

The final compensatory weights are derived by concatenation, ReLU, and Sigmoid activations, and are used to modulate the shallow visual feature

$$\boldsymbol{X}_{in}^{w} = \sigma\left(\boldsymbol{W}_{in}^{5}\left(\boldsymbol{XT}_{s}^{ma} + \mathrm{ReLU}\left(-\boldsymbol{XT}_{c}^{ma}\right)\right)\right)\boldsymbol{X}_{in} \tag{18}$$

where $\boldsymbol{W}_{in}^{5}$ denotes learnable linear parameters and $\sigma(\cdot)$ is the Sigmoid function. In the second stage, we further enhance cross-modal alignment by performing dual-branch cross-attention between the weighted visual feature $\boldsymbol{X}_{in}^{w}$ and the complete semantic text feature $\boldsymbol{t}_t$. Specifically, we use $\boldsymbol{X}_{in}^{w}$ and $\boldsymbol{t}_t$ alternately as queries to capture relationships

$$\boldsymbol{X}_{a1} = \mathrm{Att}\left(\boldsymbol{X}_{in}^{w}, \boldsymbol{t}_t, \boldsymbol{t}_t\right) \tag{19}$$

$$\boldsymbol{X}_{a2} = \mathrm{Att}\left(\boldsymbol{t}_t, \boldsymbol{X}_{in}^{w}, \boldsymbol{X}_{in}^{w}\right). \tag{20}$$

The final module output is generated by concatenating both branches and projecting them via a linear layer

$$\boldsymbol{X}_{out} = \boldsymbol{W}_{in}^{6}\mathrm{Cat}\left[\boldsymbol{X}_{a1}, \boldsymbol{X}_{a2}\right] \tag{21}$$

where $\boldsymbol{W}_{in}^{6}$ is a learnable parameter. This two-stage design enables TTIM to effectively integrate fine-grained visual details with rich semantic cues from language, providing a robust foundation for subsequent cross-modal reasoning.

D. Text-Guided Semantic Fusion Module

To further enhance multimodal semantic alignment and generate task-specific representations, the TSFM, as depicted in Fig. 3(b), is designed to achieve deep and explicit fusion between visual and textual modalities, thereby strengthening the overall representation for remote sensing tasks. The module integrates two key mechanisms: MambaTwister-based multimodal scanning and CrossAdapter-based domain adaptation. To construct explicit image-text correspondences and prevent subtle textual cues from being overwhelmed by visual features, we first compute interaction mappings between the input visual feature $\boldsymbol{X}_{in}$ and three types of textual features $\boldsymbol{t}_k$, where $k \in \{t, c, s\}$. This is achieved via matrix multiplication after linear projections, generating multimodal mappings $\boldsymbol{X}_k$

$$\boldsymbol{X}_k = \boldsymbol{W}_{in}^{7}\boldsymbol{X}_{in} \cdot \left(\boldsymbol{W}_{in}^{8}\boldsymbol{t}_k\right)^{\top}. \tag{22}$$

By concatenating the original visual feature with these multimodal mappings, a hybrid feature tensor is formed

$$\mathrm{Qube} = \mathrm{Cat}\left[\boldsymbol{X}_{in}, \boldsymbol{X}_t, \boldsymbol{X}_c, \boldsymbol{X}_s\right]. \tag{23}$$

To achieve thorough semantic fusion and ensure spatial consistency, tensor is sequentially processed by channel SSM and spatial SSM operations

$$\boldsymbol{X}_{out} = \varphi_{\mathrm{SSM_S}}\left(\varphi_{\mathrm{SSM_C}}\left(\mathrm{Qube}\right)\right). \tag{24}$$

Considering computational efficiency and the need for domain adaptation, the subsequent stage leverages only the complete text semantic feature $\boldsymbol{t}_t$ for CrossAdapter-based global fusion. The deep visual feature $\boldsymbol{X}_{in}$ and $\boldsymbol{t}_t$ are projected into a low-dimensional space via learnable parameters and ReLU

$$\boldsymbol{X}_d = \varphi_{\mathrm{ReLU}}\left(\boldsymbol{W}_{in}^{9}\boldsymbol{X}_{in}\right) \tag{25}$$

$$\boldsymbol{t}_d = \varphi_{\mathrm{ReLU}}\left(\boldsymbol{W}_{in}^{10}\boldsymbol{t}_t\right). \tag{26}$$

To enrich the visual representation, a mixture of convolutions (MOC) module is applied to $\boldsymbol{X}_d$, producing multipath features at different scales

$$\boldsymbol{X}_d^1, \boldsymbol{X}_d^2, \boldsymbol{X}_d^3 = \varphi_{\mathrm{MOC}}\left(\boldsymbol{X}_d\right). \tag{27}$$

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The outputs are concatenated and combined with the original $\boldsymbol{X}_d$ to yield the enhanced visual feature

$$\boldsymbol{X}_d^e = \mathrm{Cat}\left[\boldsymbol{X}_d^1, \boldsymbol{X}_d^2, \boldsymbol{X}_d^3\right] + \boldsymbol{X}_d. \tag{28}$$

A cross-attention mechanism is then employed to adaptively integrate the enhanced visual feature and the low-dimensional text feature, followed by an upsampling linear projection to produce the final output

$$\boldsymbol{X}_{out} = \boldsymbol{W}_{in}^{11}\left(\mathrm{Att}\left(\boldsymbol{X}_d^e, \boldsymbol{t}_d, \boldsymbol{t}_d\right) + \boldsymbol{X}_d\right). \tag{29}$$

The core visual branch within CrossAdapter, namely the MOC module, utilizes convolution kernels of varying sizes to fully exploit multiscale spatial cues

$$\boldsymbol{X}_d^1 = \varphi_{\mathrm{Conv_1}}\left(\boldsymbol{X}_d\right) \tag{30}$$

$$\boldsymbol{X}_d^2 = \varphi_{\mathrm{Conv_3}}\left(\mathrm{Cat}\left[\boldsymbol{X}_d, \boldsymbol{X}_d^1\right]\right) \tag{31}$$

$$\boldsymbol{X}_d^3 = \varphi_{\mathrm{Conv_5}}\left(\mathrm{Cat}\left[\boldsymbol{X}_d, \boldsymbol{X}_d^1, \boldsymbol{X}_d^2\right]\right) \tag{32}$$

where $\varphi_{\mathrm{Conv_k}}(\cdot)$ denotes a convolution operation with kernel size k. Through these operations, TSFM realizes fine-grained alignment and robust semantic fusion across modalities, thus enabling precise cross-modal reasoning.

IV. Experiments and Analysis

A. Dataset Description

1) RefSegRS Dataset [42]: The RefSegRS dataset is a dedicated benchmark for referring remote sensing segmentation, designed to facilitate research on integrating natural language descriptions with pixel-level segmentation in high-resolution remote sensing imagery. Built upon the Skyscapes dataset, RefSegRS contains 4420 meticulously curated image-text-mask triplets, covering 14 common object categories such as roads, buildings, and vegetation. All images are standardized to a spatial resolution of 0.13 m and resized to 512×512 pixels, ensuring consistency for model evaluation. Annotation quality is ensured through manual verification, which enhances both the accuracy of mask boundaries and the relevance of language descriptions. The dataset features significant diversity in object scale and spatial distribution, making it a robust resource for evaluating the fine-grained segmentation capability and multimodal reasoning of advanced models.

2) RRSIS-D Dataset [46]: The RRSIS-D dataset is a large-scale benchmark tailored for referring remote sensing segmentation, aimed at advancing natural language-guided fine-grained segmentation across diverse target scales and orientations in high-resolution remote sensing images. It consists of 17 402 high-quality image-text-mask triplets, with images standardized to 800×800 pixels and spatial resolutions varying from 0.5 to 30 m. RRSIS-D covers 20 object categories, including aircraft, golf courses, and highway service areas, providing rich diversity for robust model assessment. The annotation process combines the zero-shot segmentation capability of the segment anything model (SAM) for initial mask generation with manual refinement, ensuring high-precision mask boundaries and reliable textual descriptions. The dataset is split into 12 181 training, 1740 validation, and 3481 test samples, enabling standardized training and rigorous evaluation of multimodal segmentation methods.

B. Implementation Details

All experiments are conducted on workstations equipped with Intel Xeon Gold 6133 CPUs (2.50 GHz), 24 GB RAM, NVIDIA GeForce RTX 3090 GPUs, and Ubuntu 18.04. The code is implemented using PyTorch 2.0.0. Model optimization is performed with the AdamW optimizer, chosen for its effectiveness in mitigating overfitting and addressing gradient issues. Following the hyperparameter settings of FIANet [45] and employing grid search, VSPNet is trained for 200 epochs with an initial learning rate of 0.0001 and a weight decay of 0.01. The batch size is set to 4 to accommodate GPU memory.

For the loss function, we adopt the Dice loss as the sole supervision signal throughout training. Given the predicted probability p_i at pixel i and the corresponding ground-truth label y_i, with N pixels in total, the Dice coefficient is as

$$\mathrm{Dice} = \frac{2\sum_{i=1}^{N} y_i p_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} p_i}. \tag{33}$$

The Dice loss is calculated as 1 − Dice and is applied to supervise both the pseudo labels and the final predictions from the decoder. This approach encourages maximal overlap between predicted and reference regions, and is robust to the foreground-background imbalance.

C. Evaluation Metrics

To assess segmentation performance, we adopt three primary metrics: overall Intersection over Union (oIoU), mean Intersection over Union (mIoU), and precision at IoU thresholds from 0.5 to 0.9 (denoted as Pr@X), following established practices in the field. The oIoU metric, defined as the ratio of the total intersection area to the total union area over the entire test set, is formulated as

$$\mathrm{oIoU} = \frac{\sum_{i=1}^{N} |P_i \cap G_i|}{\sum_{i=1}^{N} |P_i \cup G_i|} \tag{34}$$

where P_i and G_i denote the predicted and ground-truth foreground pixel sets for the ith image, respectively. This metric emphasizes the segmentation accuracy of large-scale objects by assigning greater weight to larger regions. In contrast, mIoU computes the average Intersection over Union across all test samples, treating each sample equally regardless of object size. For each class c, the IoU is defined as

$$\mathrm{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{35}$$

where TP_c, FP_c, and FN_c represent the true positive, false positive, and false negative counts for class c. The mIoU is then calculated as follows:

$$\mathrm{mIoU} = \frac{1}{K}\sum_{c=1}^{K} \mathrm{IoU}_c \tag{36}$$

with K denoting the number of classes. mIoU is widely used due to its balanced treatment of all categories, but may underrepresent minority classes without additional weighting.

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TABLE I

QUANTITATIVE COMPARISON OF RIS METHODS ON THE REFSEGRS DATASET. RESULTS ARE REPORTED FOR VARIOUS EVALUATION METRICS, INCLUDING PR@X, OIOU, MIOU. THE BEST PERFORMANCE FOR EACH METRIC IS HIGHLIGHTED IN BOLD

Method	Publication	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
LSTM-CNN [55]	ECCV'2016	15.69	10.57	5.17	1.10	0.28	53.83	24.76
ConvLSTM [56]	CVPR'2018	31.21	23.39	15.30	7.59	1.10	66.12	43.34
CMSA [57]	CVPR'2019	28.07	20.25	12.71	5.61	0.83	64.53	41.47
BRINet [58]	CVPR'2020	22.56	15.74	9.85	3.52	0.50	60.16	32.87
LAVT [38]	CVPR'2022	70.23	55.53	30.05	14.42	4.07	76.21	57.30
CrossVLT [59]	TMM'2023	71.16	58.28	34.51	16.35	5.06	77.44	58.84
RMISN [46]	CVPR'2024	71.60	55.97	31.87	11.72	1.93	71.73	57.78
LGCE [60]	TGRS'2024	76.55	67.03	44.85	19.04	5.67	77.62	61.90
FIANet [45]	TGRS'2024	84.09	77.05	61.86	33.41	7.10	78.32	68.67
CADFormer [48]	JSTARS'2025	79.82	65.12	50.23	20.19	3.41	73.12	65.41
RefSegformer [61]	TIP'2024	50.25	28.01	17.83	9.19	2.48	70.13	47.12
Remamber [40]	ECCV'2024	68.33	59.24	50.32	32.17	4.87	76.35	63.16
DANet [62]	ACM MM'2024	66.57	57.11	48.16	25.16	4.33	73.19	64.12
ReferSAM [63]	TCSVT'2025	76.44	75.13	60.62	35.26	7.01	78.51	66.96
Enti-TwistNet [64]	TGRS'2025	60.10	70.67	57.02	30.12	7.34	74.00	64.67
Ours	—	**85.12**	**78.59**	**63.01**	**35.42**	**9.23**	**79.60**	**69.59**

D. Comparison With State-of-the-Art Methods

To rigorously evaluate the effectiveness of our proposed multimodal fusion approach, we compare it against fifteen representative methods from the field of remote sensing RIS, covering the main technological milestones of the past decade. The selected baselines span early hybrid architectures (e.g., LSTM-CNN [55], RNN [56], LSCM [65], ConvLSTM [56]), attention-based models (e.g., CMSA [57], CMPC [66], CMPC+ [67], BRINet [58]), and recent advanced multimodal fusion frameworks (e.g., LAVT [38], CrossVLT [59]). All compared methods are state-of-the-art, ensuring both comprehensiveness and timeliness. Moreover, the latest approaches introduced between 2024 and 2025, including RMISN [46], LGCE [60], FIANet [45], CADFormer [48], RefSegformer [61], Remamber [40], DANet [62], ReferSAM [63], and Enti-TwistNet [64] are incorporated as strong baselines to enable a systematic and fair performance comparison of our method. The specific analysis is as follows:

1) Performance Comparison on RefSegRS Dataset: To comprehensively evaluate our proposed multimodal fusion framework for referring segmentation in remote sensing images, we conduct systematic quantitative comparisons on the RefSegRS dataset. As summarized in Table I, our method consistently outperforms all competing approaches across every evaluation metric. Specifically, our approach achieves 85.12% on Pr@0.5, surpassing the previous state-of-the-art FIANet (84.09%) by 1.03%. On the more challenging Pr@0.6 metric, we obtain 78.59%, a 1.54% improvement over FIANet. At the stringent Pr@0.9 threshold, our method achieves 9.23%, outperforming the next best method by 2.13%, highlighting its exceptional capability for precise segmentation. Regarding overall segmentation quality, our approach attains 79.60% on oIoU and 69.59% on mIoU, establishing new state-of-the-art results. Compared to early approaches, such as LSTM-CNN, the improvement in mIoU reaches 44.83%, underscoring the effectiveness of modern multimodal fusion. Relative to recent advanced methods, our method outperforms CADFormer by 4.18% and FIANet by 0.92% in mIoU, demonstrating the importance of tailored fusion strategies that leverage the strengths of diverse architectures. A fine-grained category analysis (Table II) further illustrates the superiority of our method, which achieves the best results in 8 out of 14 categories, including road, car, and bus.

TABLE II

QUANTITATIVE COMPARISON OF RIS METHODS ON THE REFSEGRS DATASET. RESULTS ARE REPORTED FOR VARIOUS EVALUATION METRICS, INCLUDING PR@X, OIOU, MIOU. THE BEST PERFORMANCE FOR EACH METRIC IS HIGHLIGHTED IN BOLD

Category	LAVT	RMSIN	LGCE	FIANeT	Ours
Road	70.33	66.67	74.03	74.13	**74.76**
Vehicle	57.02	58.66	63.05	**70.15**	69.33
Car	55.13	57.63	60.79	68.55	**70.12**
Van	38.55	47.09	41.60	61.06	**61.62**
Building	81.92	76.84	**81.99**	81.34	81.36
Truck	53.07	51.92	62.69	**74.48**	72.28
Trailer	44.56	61.65	49.82	**74.92**	74.77
Bus	52.93	60.20	45.36	72.40	**73.08**
Road Marking	5.74	18.60	6.66	22.85	**23.18**
Bikeway	50.26	50.35	54.23	61.16	**63.36**
Sidewalk	57.35	49.12	61.68	**62.90**	60.10
Tree	57.01	49.82	67.68	83.75	**84.25**
Low Vegetation	41.08	43.73	43.68	**44.84**	43.16
Impervious Surface	81.51	76.55	83.18	81.53	**82.42**
Average	53.32	54.92	56.89	**66.72**	66.70

Notably, our framework excels in vehicle-related classes, achieving 70.12% for cars (1.57% higher than the best baseline) and 61.62% for vans (a substantial 20.02% improvement over LGCE). For challenging thin structures such as road markings, our method attains 23.18%, outperforming FIANet by 0.33%. While marginally trailing in a few categories (e.g., building), our average performance across all categories remains competitive at 66.70%, nearly matching the best result (66.72%). These results collectively indicate that remote sensing images pose significant challenges due to

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TABLE III

QUANTITATIVE COMPARISON OF RIS METHODS ON THE RRSIS-D DATASET. RESULTS ARE REPORTED FOR VARIOUS EVALUATION METRICS, INCLUDING PR@X, OIOU, MIOU. THE BEST PERFORMANCE FOR EACH METRIC IS HIGHLIGHTED IN BOLD

Method	Publication	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
RRN [56]	CVPR'2018	51.07	42.11	32.77	21.57	6.37	66.43	45.64
CMSA [57]	CVPR'2019	55.32	46.45	37.43	25.39	8.15	69.39	48.54
LSCM [65]	ECCV'2020	56.02	46.25	37.70	25.28	8.27	69.05	49.92
CMPC [66]	CVPR'2020	55.83	47.40	36.94	25.45	9.19	69.22	49.24
BRINet [58]	CVPR'2020	56.90	48.77	39.12	27.03	8.73	69.88	49.65
CMPC+ [67]	TPAMI'2021	57.65	47.51	36.97	24.33	7.78	68.64	50.24
LAVT [38]	CVPR'2022	66.93	60.99	51.71	39.79	23.99	76.58	59.05
CrossVLT [59]	TMM'2023	70.38	63.83	52.86	42.11	25.02	76.32	61.00
LGCE [60]	TGRS'2024	69.41	63.06	53.46	41.22	24.27	76.24	61.02
RMISN [46]	CVPR'2024	71.96	65.76	55.16	42.03	25.02	76.50	62.27
FIANet [45]	TGRS'2024	74.46	66.96	56.31	42.83	24.13	76.91	64.01
CADFormer [48]	JSTARS'2025	74.20	**67.62**	**55.59**	42.37	23.59	77.26	63.77
RefSegformer [61]	TIP'2024	66.59	59.58	49.93	33.78	23.30	76.40	58.99
Remember [40]	ECCV'2024	68.33	60.24	50.32	32.17	20.63	75.35	61.26
DANet [62]	ACM MM'2024	70.19	61.21	44.34	30.28	18.42	74.66	57.34
ReferSAM [63]	TCSVT'2025	72.21	62.98	49.17	33.16	23.97	75.49	62.01
Enti-TwistNet [64]	TGRS'2025	74.88	60.10	54.76	39.21	24.26	77.02	64.45
Ours	—	**75.06**	67.58	55.42	**44.61**	**25.43**	**77.81**	**64.69**

large object scale variation, spatial diversity, and complex contexts. The proposed hybrid feature extraction architecture (CNN-Transformer-Mamba) enables the model to capture both fine-grained textures and global semantic information, resulting in superior adaptability to the complexities of remote sensing data compared to single-architecture baselines. Overall, our study provides an effective solution to multiscale object segmentation and semantic alignment in remote sensing imagery, advancing the state of the art in referring segmentation.

2) Performance Comparison of RRSIS-D Dataset: To further validate the generalization and robustness of the proposed VSPNet, we conduct comprehensive quantitative experiments on the large-scale RRSIS-D dataset, which contains 20 object categories and significantly more samples than RefSegRS. As shown in Table III, our method establishes new state-of-the-art results across all evaluation metrics. Specifically, our approach achieves 75.06% on Pr@0.5, outperforming the previous best methods, FIANet and CADFormer, by 0.60% and 0.86%, respectively. Under stricter thresholds, our framework achieves 44.61% on Pr@0.8 and 25.43% on Pr@0.9, demonstrating clear superiority in high-precision segmentation scenarios. In terms of segmentation quality, our method attains 64.69% in mIoU, exceeding the second-best method FIANet (64.01%) by 0.68%. Compared to the early baseline RRN, this constitutes a substantial 19.05% improvement, highlighting the effectiveness of multimodal fusion for remote sensing image segmentation. Furthermore, our approach achieves the highest oIoU at 77.81%, further validating its ability to maintain global segmentation integrity.

A fine-grained category-wise analysis, presented in Table IV, shows that our method achieves the best performance in 12 out of 19 categories, including highly structured scenes such as airport, stadium, and athletic field. For challenging categories like bridge and dam, our approach achieves 40.02% and 72.47%, outperforming FIANet by 2.08% and 1.16%, respectively. These results highlight the model's capacity to accurately capture complex semantics and spatial relationships in diverse geospatial contexts. The average category performance reaches 67.12%, which significantly surpasses all comparative methods. Overall, these results confirm the proposed VSPNet framework's superior generalization and robustness on the challenging, large-scale RRSIS-D dataset. The consistently strong performance, especially under high-threshold precision metrics, demonstrates the effectiveness of our hierarchical multimodal fusion strategy. Notably, the marked improvements on complex categories such as bridge

TABLE IV

QUANTITATIVE COMPARISON OF RIS METHODS ON THE RRSIS-D DATASET. RESULTS ARE REPORTED FOR VARIOUS EVALUATION METRICS, INCLUDING PR@X, OIOU, AND MIOU. THE BEST PERFORMANCE FOR EACH METRIC IS HIGHLIGHTED IN BOLD

Category	LAVT	RMSIN	LGCE	FIANet	Ours
Airport	66.44	68.08	68.11	68.66	**68.90**
Golf Field	56.53	56.11	56.43	57.07	**58.42**
Expressway Service Area	76.08	76.68	77.19	**77.35**	76.32
Baseball Field	68.56	66.93	70.93	70.44	**71.32**
Stadium	81.77	83.09	84.90	84.87	**85.16**
Ground Track Field	81.84	81.91	82.54	82.00	**83.40**
Storage Tank	71.33	73.65	73.33	**76.99**	76.02
Basketball Court	70.71	72.26	74.37	**74.86**	74.78
Chimney	65.54	68.42	68.44	68.41	**68.99**
Tennis Court	74.98	76.68	75.63	78.48	**79.42**
Overpass	66.17	**70.14**	67.67	70.01	68.67
Train Station	57.02	62.67	58.19	61.30	**63.45**
Ship	63.47	64.64	63.48	65.96	**66.73**
Expressway Toll Station	63.01	**65.71**	61.63	64.82	63.19
Dam	61.61	68.70	64.54	71.31	**72.47**
Harbor	60.05	60.40	60.47	62.03	**62.48**
Bridge	30.48	36.74	34.24	37.94	**40.02**
Vehicle	42.60	47.63	43.12	**49.66**	48.96
Windmill	35.32	41.99	40.76	**46.72**	46.51
Average	62.44	65.13	64.12	66.46	**67.12**

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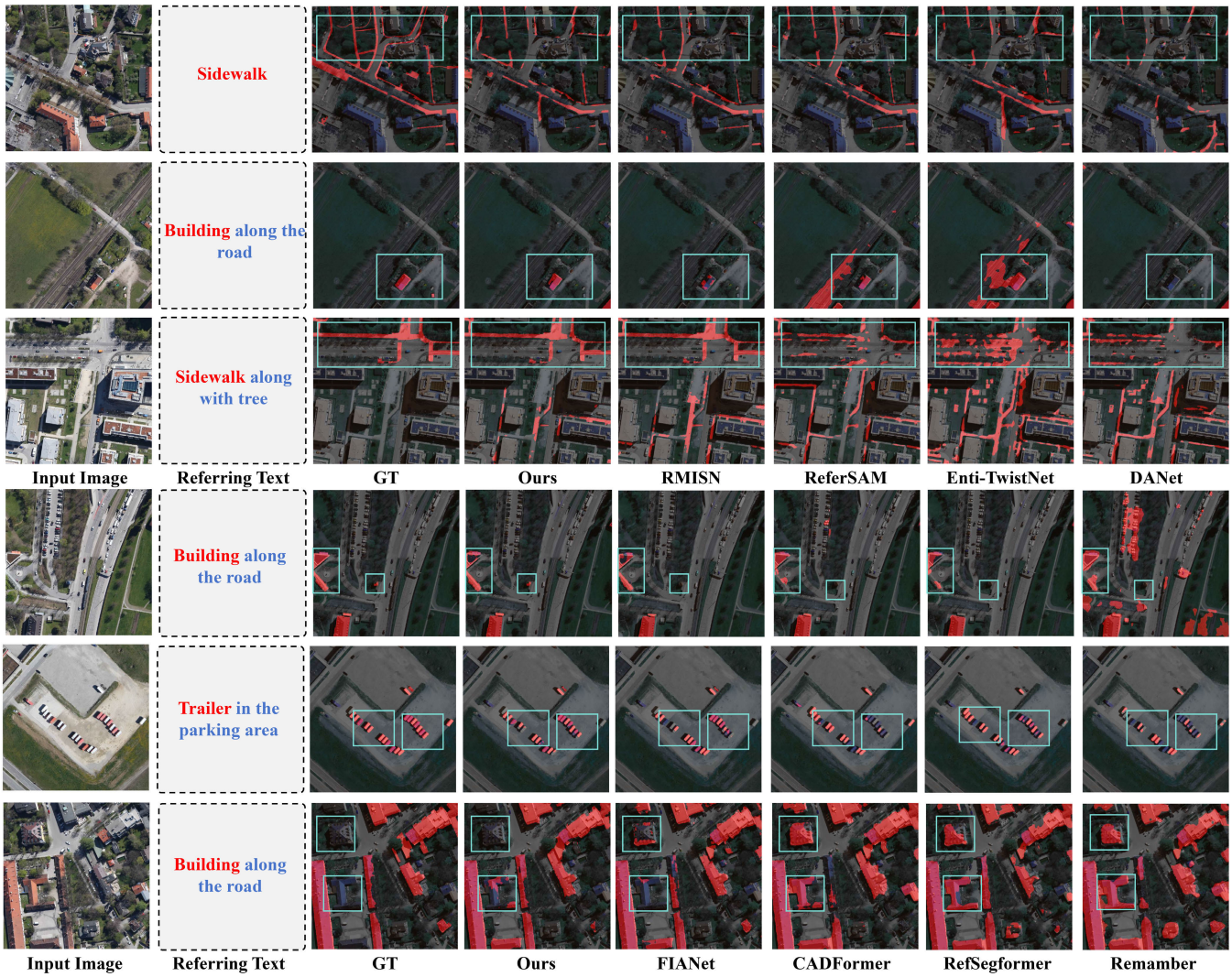


Fig. 4. Qualitative comparison of segmentation results on the RefSegRS dataset. The visualization presents representative urban scenarios, including vehicles, roads, and buildings, where the results of the proposed VSPNet are compared against state-of-the-art methods. The examples highlight the superior localization accuracy, semantic consistency, and robustness of VSPNet in handling complex scenes.

and dam further validate the synergy between the TTIM and TSFM modules in decoupling and aligning spatial relationships from linguistic cues with global visual context. This work not only sets a new performance benchmark for referring segmentation in remote sensing but also provides a robust methodological foundation for multimodal understanding of complex geospatial scenes.

3) Scenario Comparison on RefSegRS and RRSIS-D Datasets: To further validate the perceptual effectiveness of our hierarchical vision-language fusion framework, we conducted systematic qualitative analyses on both the RefSegRS and RRSIS-D datasets. As illustrated in Fig. 4, we selected representative urban scenarios from RefSegRS, including parking lots, road networks, and building areas, to compare our method against leading approaches such as FIANet and ReferSAM. Benefiting from the multidimensional representation capability of the CMTFE, our framework effectively integrates local texture details with global contextual information, substantially improving segmentation precision. In vehicle-related detection tasks, our method demonstrates clear advantages. For example, when segmenting "sidewalk area," FIANet suffers from significant missed detections, while CADFormer exhibits noticeable localization errors. By enhancing shallow feature discriminability through the TTIM, our approach aligns closely with the ground truth, particularly improving the localization of low-contrast vehicles and confirming the effectiveness of our multimodal alignment strategy. For complex structures such as "buildings along the road," the TSFM enables fine-grained alignment between deep semantics and linguistic cues, accurately distinguishing buildings adjacent to different types of roads. This overcomes common issues of overlap and false detection seen in other methods. In scenarios with challenging visual conditions, such as "building along the road," our approach, leveraging multiscale CMTFE and the contextual modeling of TSFM, achieves complete contour extraction and significantly outperforms comparative baselines that struggle with missed detections. Multiobject scenarios, such as parking lots containing trailers, further highlight the robustness of our method. The hierarchical fusion mechanism enables effective disentanglement of object semantics and spatial relations from

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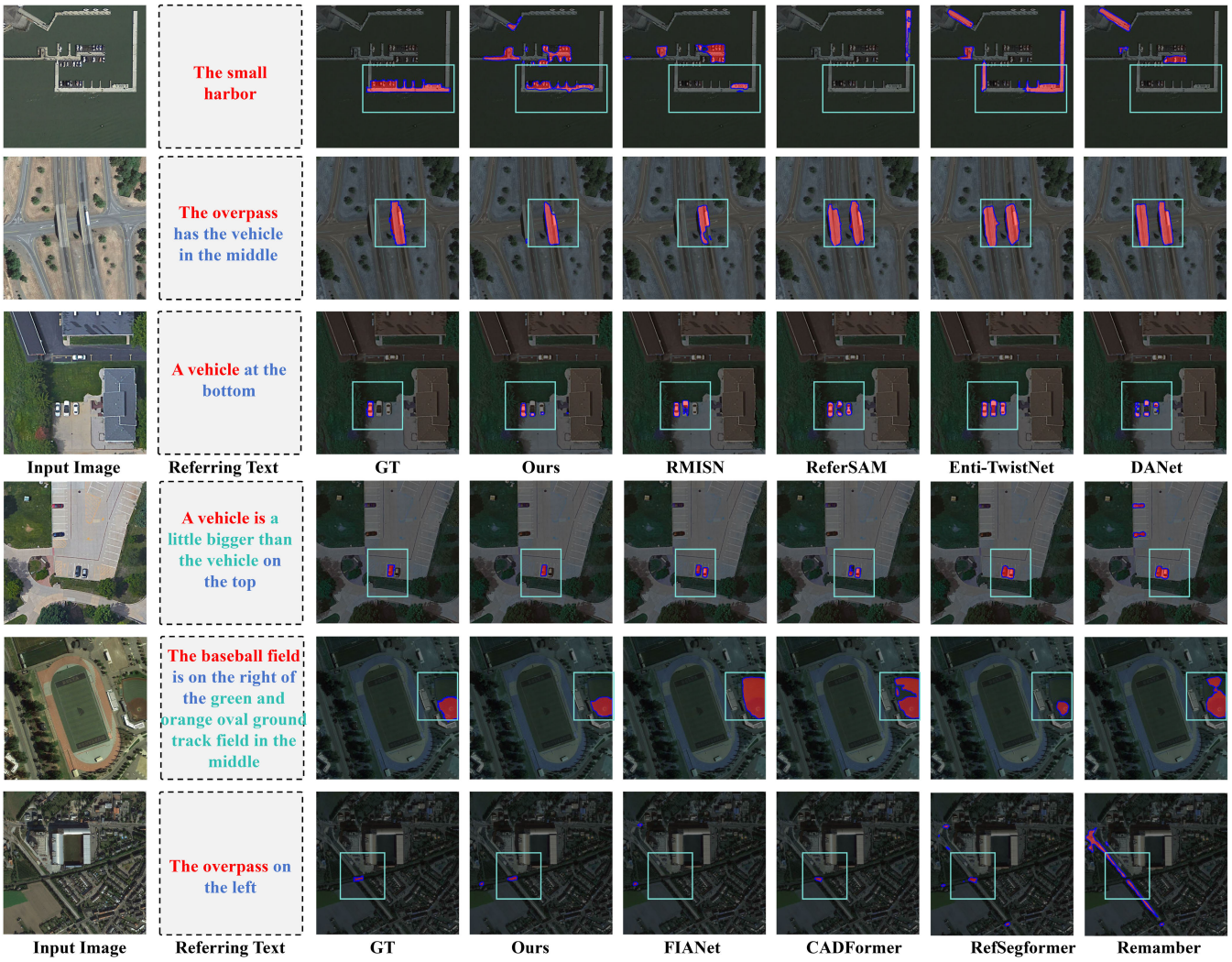


Fig. 5. Qualitative comparison of segmentation results on the RRSIS-D dataset. The visualization covers a variety of challenging remote sensing scenes, such as sparse vehicles, green golf fields, and shadowed airports. The proposed VSPNet is compared with leading methods, demonstrating its ability to produce precise boundaries, and maintain semantic consistency.

textual descriptions, thereby distinguishing between easily confused targets and greatly enhancing discriminative capability. The end-to-end multimodal interaction design ensures precise localization and high reliability even in highly complex scenes. As shown in Fig. 5, qualitative results on the RRSIS-D dataset further demonstrate that VSPNet's hierarchical multimodal fusion architecture substantially strengthens the model's ability to synergistically model subtle spatial relationships from language and contextual dependencies from visual scenes. In challenging scenarios, such as sparse vehicles, green golf fields, and shadowed airports, our method consistently yields segmentation results with precise boundaries and clear semantic alignment, effectively addressing issues such as false positives, missed detections, and blurred boundaries prevalent in other approaches. This superior performance is attributed to the TTIM enhancement of visual details and the TSFM precise alignment of deep semantic information, whose collaboration ensures strong adaptability to large-scale variations and complex spatial relationships in remote sensing imagery. Overall, the proposed VSPNet framework achieves exceptional segmentation accuracy and robustness across a wide variety of complex remote sensing scenarios, setting a new benchmark for high-difficulty geospatial referring segmentation tasks.

E. Ablation Studies

To systematically evaluate the effectiveness of each core component in VSPNet, we conduct ablation experiments on the RefSegRS and RRSIS-D datasets. The baseline adopts the classical image-text alignment scheme of LAVT [38].

All results are summarized in Tables V and VI, with qualitative heatmap visualizations shown in Fig. 6.

1) Effectiveness of CMTFE: The hybrid feature extractor structure, named CMTFE is designed to address the inherent scale variation and context sparsity in remote sensing imagery. By integrating multiscale convolutional and transformer branches, CMTFE is capable of simultaneously capturing fine-grained local structures as well as long-range dependencies. Quantitatively, as displayed in Tables V and VI, the incorporation of CMTFE into the baseline brings a

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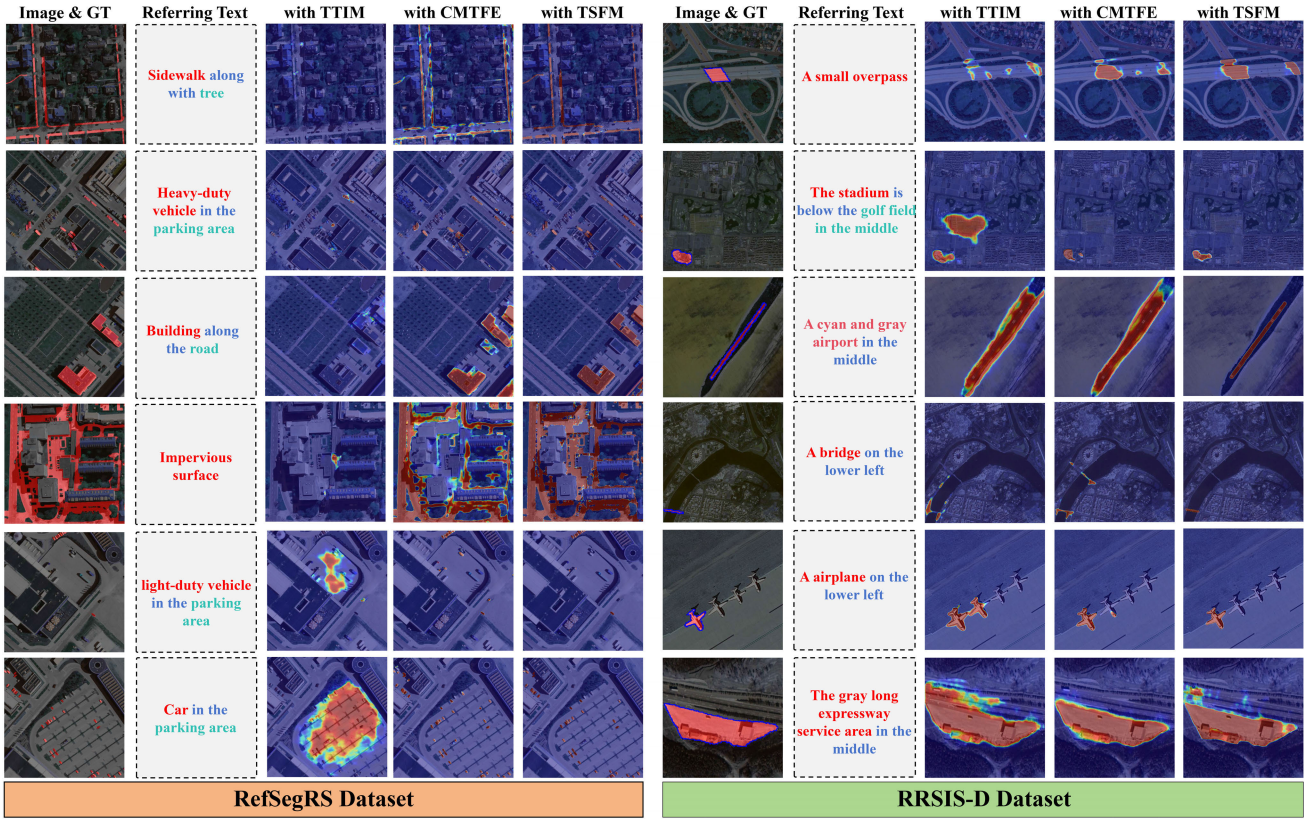


Fig. 6. Qualitative visualization of feature heatmaps from ablation experiments on the RefSegRS and RRSIS-D datasets. For each scenario, heatmaps are presented after sequentially integrating TTIM, CMTFE, and TSFM modules.

TABLE V

ABLATION RESULTS OF THE PROPOSED VSPNET ON THE REFSEGRS AND RRSIS-D DATASETS. EACH ROW SHOWS THE INCREMENTAL EFFECT OF ADDING CMTFE, TTIM, AND TSFM MODULES TO THE BASELINE

Method	Refseg							RRSIS-D						
	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
Baseline	80.13	69.13	44.67	18.37	2.75	75.49	66.30	72.56	56.42	48.59	31.99	9.36	72.11	60.12
+ CMTFE	81.41	72.94	53.16	25.62	3.52	76.50	66.82	73.00	55.19	51.39	36.98	14.41	73.34	60.97
+ CMTFE + TIIM	82.98	79.22	57.92	37.16	6.19	78.01	67.91	73.87	65.17	57.29	40.13	20.78	75.91	62.53
+ CMTFE + TIIM + TSFM	85.12	78.59	63.01	35.42	9.23	79.60	69.59	75.06	67.58	55.42	44.61	25.43	77.81	64.69

TABLE VI

ABLATION STUDY ON DIFFERENT MODULE COMBINATIONS IN VSPNET ON THE REFSEGRS AND RRSIS-D DATASETS. THIS TABLE COMPARES VARIOUS PAIRWISE AND FULL COMBINATIONS OF THE CMTFE, TTIM, AND TSFM MODULES

Method	Refseg							RRSIS-D						
	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU	Pr@0.5	Pr@0.6	Pr@0.7	Pr@0.8	Pr@0.9	oIoU	mIoU
+ CMTFE + TSFM	78.32	74.94	55.92	28.28	7.14	76.19	67.72	74.01	64.19	53.17	32.13	21.46	76.00	62.06
+ TIIM + TSFM	84.02	73.10	56.90	32.29	7.86	77.29	68.00	72.99	66.14	52.43	38.71	19.46	74.97	61.45
+ CMTFE + TIIM	82.98	79.22	57.92	37.16	6.19	78.01	67.91	73.87	65.17	57.29	40.13	20.78	75.91	62.53
Ours	85.12	78.59	63.01	35.42	9.23	79.60	69.59	75.06	67.58	55.42	44.61	25.43	77.81	64.69

consistent improvement across all metrics: mIoU increases from 66.30% to 66.82% on RefSegRS, and from 60.12% to 60.97% on RRSIS-D. More notably, CMTFE significantly boosts the high-threshold Pr@0.9 metric on RefSegRS from 2.75% to 3.52% and on RRSIS-D from 9.36% to 14.41%. These improvements highlight CMTFE's ability to provide richer and more robust feature representations, especially for small or low-contrast objects that are otherwise difficult to segment. Qualitatively, as shown in the heatmap of Fig. 6, adding CMTFE results in more concentrated and confident activation on target regions, while suppressing irrelevant background noise. The module enables the network to better localize

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and delineate object contours, laying a solid foundation for downstream cross-modal alignment. Practically, the enhanced multiscale perception introduced by CMTFE is crucial for handling the diverse spatial distribution and appearance variability typical in remote sensing scenarios.

2) Effectiveness of TTIM: The TTIM is proposed to achieve fine-grained cross-modal alignment by directly injecting decoupled textual cues into shallow visual representations. This design is motivated by the observation that accurate boundary localization and object distinction in referring segmentation often require explicit guidance from language at the pixel level. Upon introducing TTIM (see Tables V and VI), significant gains are observed in both average and high-precision metrics. On RRSIS-D, Pr@0.9 improves from 14.41% to 20.78%, and on RefSegRS, Pr@0.8 rises from 25.62% to 35.42%. The module also lifts mIoU to 67.91% and 62.53% on the two datasets, respectively. These results demonstrate that TTIM enables the model to more accurately resolve ambiguous boundaries and distinguish between closely situated or visually similar objects based on linguistic cues. From the visualization perspective (Fig. 6), TTIM significantly sharpens the model's focus on the target regions specified by the referring text, with heatmaps clearly outlining object edges and suppressing confusion from the surrounding context. In practice, TTIM's fine-grained alignment mechanism greatly benefits scenarios involving dense object distributions or overlapping targets, as it leverages the discriminative power of natural language to guide the segmentation process at the detail level.

3) Effectiveness of TSFM: The TSFM serves as the final stage of hierarchical fusion, aiming to deeply integrate high-level visual semantics with comprehensive textual understanding. Based on the Mamba architecture, TSFM allows for global context modeling and precise alignment of complex spatial relationships described by language. The full integration of TSFM into the pipeline yields the most substantial performance improvements (Tables V and VI): mIoU reaches 69.59% on RefSegRS and 64.69% on RRSIS-D, while Pr@0.9 achieves 9.23% and 25.43%, respectively, the highest among all configurations. TSFM's impact is also evident in the substantial reduction of misclassification and false positives, particularly for challenging and composite referring expressions. The qualitative results in Fig. 6 show that TSFM further enhances the model's ability to capture the overall structure and semantic consistency of target objects. The heatmaps become more coherent and tightly aligned with object boundaries, even in scenes with heavy occlusion or complex spatial relationships. In practical terms, TSFM's global semantic fusion is essential for interpreting long and complex referring expressions, as well as for maintaining segmentation integrity across large-scale or contextually rich scenes, challenges that are common in remote sensing image analysis.

F. Failure Case Analysis

While VSPNet achieves competitive performance, certain limitations persist on both the RefSegRS and RRSIS-D datasets.

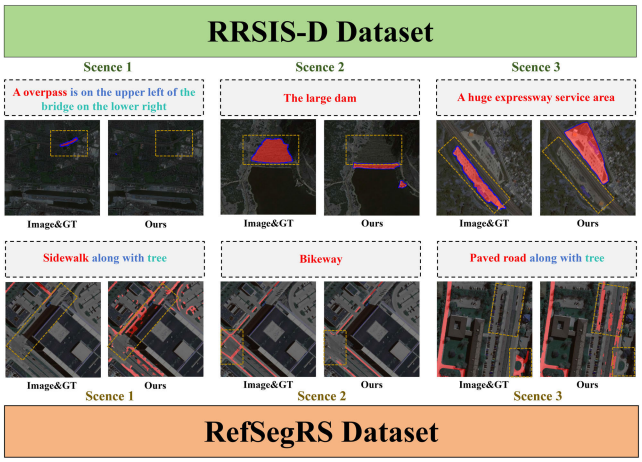


Fig. 7. Visualization of representative failure cases of VSPNet on the RefSegRS and RRSIS-D datasets.

As depicted in Fig. 7, representative failure cases reveal two primary challenges: On RefSegRS, VSPNet occasionally overemphasizes spatial and contextual cues within referring expressions, leading to the erroneous inclusion of visually similar but semantically irrelevant regions, such as adjacent sidewalks or tree areas. This issue largely stems from the CLIP text encoder, which, being pretrained on natural scenes, may inadequately capture domain-specific spatial semantics and subtle contextual distinctions relevant to remote sensing. Consequently, semantic boundaries between target and non-target regions can become ambiguous. On RRSIS-D, VSPNet struggles with the segmentation of slender or elongated objects described by complex expressions, especially when spatial relationships are intricate or when similar objects co-occur. This reflects the persistent semantic gap between nuanced linguistic descriptions and multiscale visual representations in remote sensing imagery. These observations underscore the need for enhanced cross-modal discrimination and fine-grained semantic reasoning. Future work will explore domain-adaptive pretraining or fine-tuning of text encoders to better capture remote sensing-specific semantics and to improve the model's robustness in challenging referring segmentation scenarios.

V. Conclusion

In this work, we present VSPNet, a novel end-to-end framework for remote sensing RIS. Distinct from existing approaches, VSPNet introduces a hybrid feature extraction framework that hierarchically combines CNN, Mamba, and Transformer architectures. This design enables simultaneous capture of fine-grained local textures and complex semantic context, effectively addressing the multiscale and semantic diversity of remote sensing imagery. To bridge the vision-language gap, we propose the TTIM for shallow feature enhancement and the TSFM for deep semantic alignment. This hierarchical multimodal fusion strategy empowers VSPNet with robust and flexible reasoning from coarse localization to fine segmentation. Extensive experiments on RefSegRS and RRSIS-D benchmarks demonstrate that VSPNet achieves state-of-the-art performance across all key metrics. Ablation

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studies further confirm the indispensable and complementary roles of each module, while qualitative results highlight strong generalization in complex remote sensing scenarios. Future work will focus on enhancing the scalability and generalization of VSPNet, including the development of more universal and robust foundation models for remote sensing image segmentation.

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